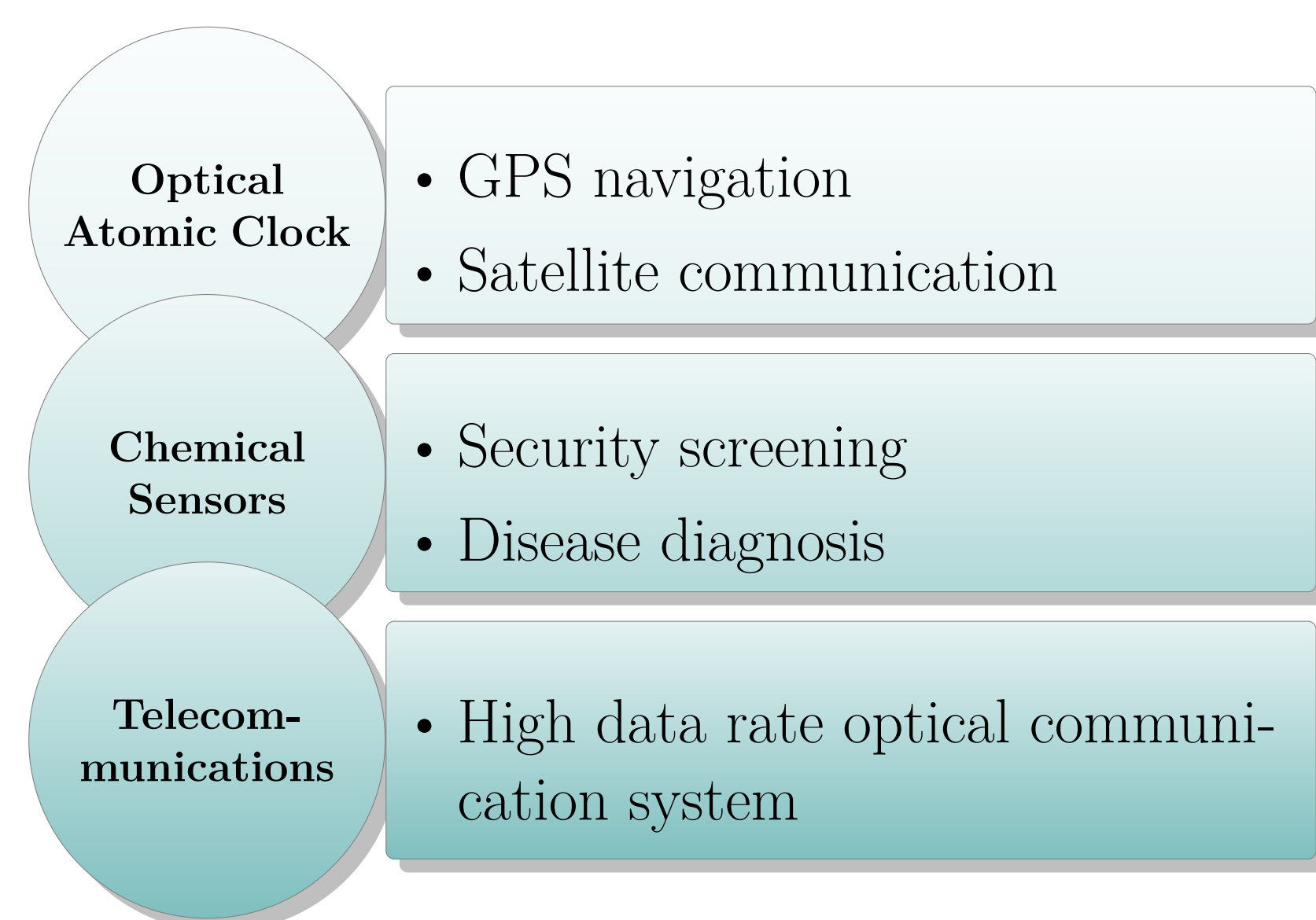


Mode-locked Fiber Lasers

Generate a regular train of short, high-energy light pulses by balancing

- Gain and Loss
- Dispersion and Nonlinearity.

Applications



Lumped Model of Stretched Pulse Laser^[1]

t = evolution parameter and x = time.

1. Fast Saturable Absorber:

$$\psi_{\text{out}} = \left(1 - \frac{l_0}{1 + |\psi_{\text{in}}|^2/P_{\text{sat}}}\right) \psi_{\text{in}}.$$

2. Single Mode Fiber (NLS Equation):

$$\partial_t \psi = \left[-\frac{i\beta_{\text{SMF}}}{2} \partial_x^2 + i\gamma|\psi|^2\right] \psi.$$

3. Fiber Amplifier:

$$\partial_t \psi = \left[-\frac{i\beta_{\text{FA}}}{2} \partial_x^2 + i\gamma|\psi|^2 + \frac{g(\psi)}{2} \left(1 + \frac{1}{\Omega_g^2} \partial_x^2\right)\right] \psi,$$

where

$$g(\psi) = \frac{g_0}{1 + E(\psi)/E_{\text{sat}}}.$$

4. Dispersion Compensation Fiber:

$$\partial_t \psi = -\frac{i\beta_{\text{DCF}}}{2} \partial_x^2 \psi.$$

5. Output Coupler:

$$\psi_{\text{out}} = \frac{1}{\sqrt{2}} \psi_{\text{in}}.$$

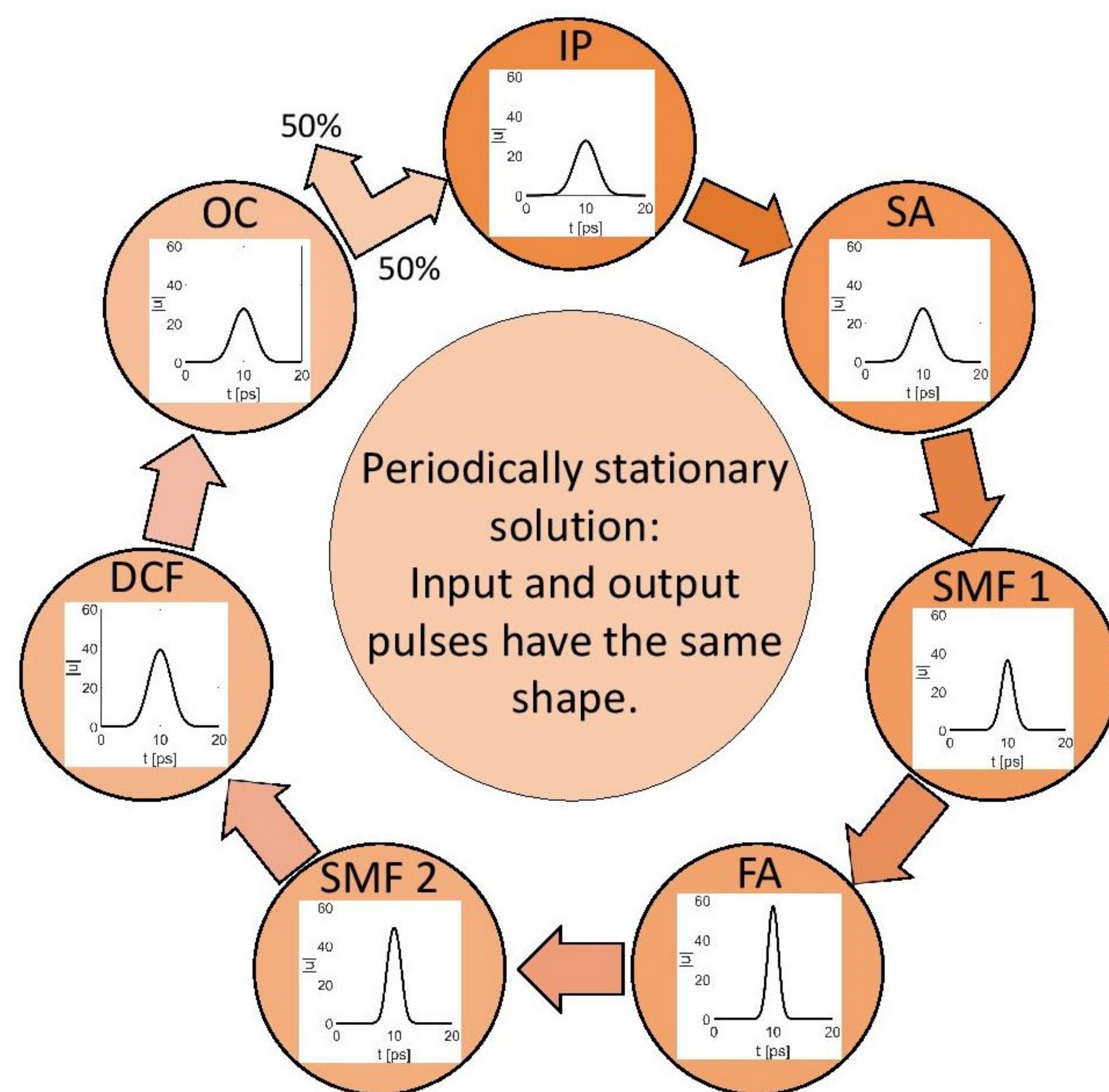


Fig. 1: Periodically stationary pulse obtained using nonlinear PDE solver.

If **periodically stationary solution (breather)** exists, then

$$\psi_{\text{L}} = \mathcal{R}(\psi_0) = e^{i\phi_0} \psi_0, \quad \text{for some } \phi_0.$$

Here, $\mathcal{R}$ is the composition of the evolution operators for each component.

Slow Saturable Absorber

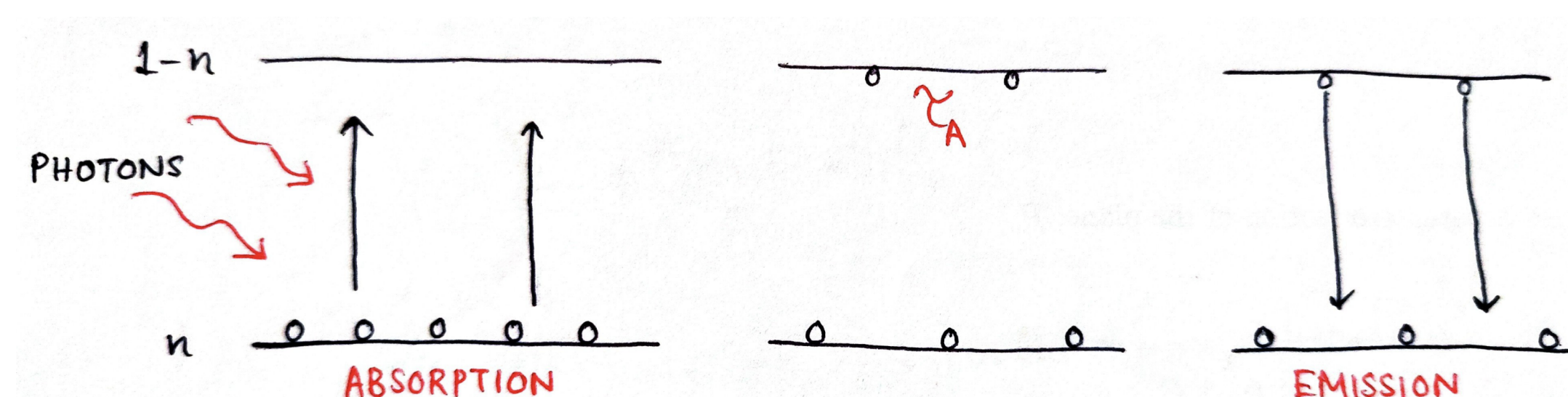


Fig. 2: Two-level energy model of SESAM.

To model realistic saturation dynamics we replace the fast saturable absorber^[2] by a slow saturable absorber (SESAM) with a non-zero response time τ_A . This transfer function for this SESAM is given by

$$\psi_{\text{out}} = \mathcal{P}^{\text{SESAM}}(\psi_{\text{in}}(\mathbf{x})) = \exp(-\rho n(\mathbf{x})) \psi_{\text{in}}.$$

Here n satisfies

$$\frac{dn}{dx} = \frac{1-n}{\tau_A} - \frac{|\psi(\mathbf{x})|^2}{w_A} n, \quad n(\mathbf{x} = -\infty) = 1.$$

and is given by

$$n(x) = \exp\left(-\frac{w(x) - w(x_0)}{w_A} - \frac{x - x_0}{\tau_A}\right) \left[1 + \frac{1}{\tau_A} \int_{x_0}^x \exp\left(\frac{w(x') - w(x_0)}{w_A} + \frac{x' - x_0}{\tau_A}\right) dx'\right],$$

where

$$w(x) = \int_{-\infty}^x |\psi(x')|^2 dx'.$$

Stability Analysis

To determine the stability of the pulse, we compute the spectrum of the monodromy operator $\mathcal{M}$, which is the linearization of the roundtrip operator, $\mathcal{R}$ ^[3].

• Linearization in SESAM:

The linearization of the transfer function in SESAM is given by

$$\mathbf{u}_{\text{out}}(x) = \mathcal{U}^{\text{SESAM}} \mathbf{u}_{\text{in}} = \exp(-\rho n(\psi_{\text{in}})) [I - \rho \Delta n(\psi_{\text{in}}) \psi_{\text{in}}] (\mathbf{u}_{\text{in}}).$$

Here $\Delta n(\psi_{\text{in}})(\mathbf{u}_{\text{in}})$ is the linearization of n

$$\Delta n(\psi_{\text{in}})(\mathbf{u}_{\text{in}}) = \mathbf{f}(\mathbf{x}) \left(\int_{x_0}^x \psi_{\text{in}}^T(\mathbf{x}') \mathbf{u}_{\text{in}}(\mathbf{x}') d\mathbf{x}' \right) + \mathbf{g}(\mathbf{x}) \int_{x_0}^x \mathbf{h}(\mathbf{x}') \left(\int_{x_0}^x \psi_{\text{in}}^T(\mathbf{s}) \mathbf{u}_{\text{in}}(\mathbf{s}) d\mathbf{s} \right) d\mathbf{x}',$$

where

$$f(x) = -\tau_A g(x) \left[1 + \frac{1}{\tau_A} \int_{x_0}^x h(x') dx'\right] \quad \text{and} \\ g(x) = \frac{2}{w_A \tau_A h(x)} \quad \text{for} \\ h(x') = \exp\left(\frac{w(x') - w(x_0)}{w_A} + \frac{x' - x_0}{\tau_A}\right).$$

• Adjoint state method

Finding breathers by optimization involves computation of the adjoint of $\mathcal{M}$ ^[2]. For SESAM,

$$(\mathcal{U}^{\text{SESAM}})^*(\mathbf{v}) = \exp(-\rho n) \mathbf{v} - \rho T^*(\exp(-\rho n) \psi^T \mathbf{v}),$$

where $T(\mathbf{u}) = \Delta n(\psi)(\mathbf{u})$.

Numerical Simulations

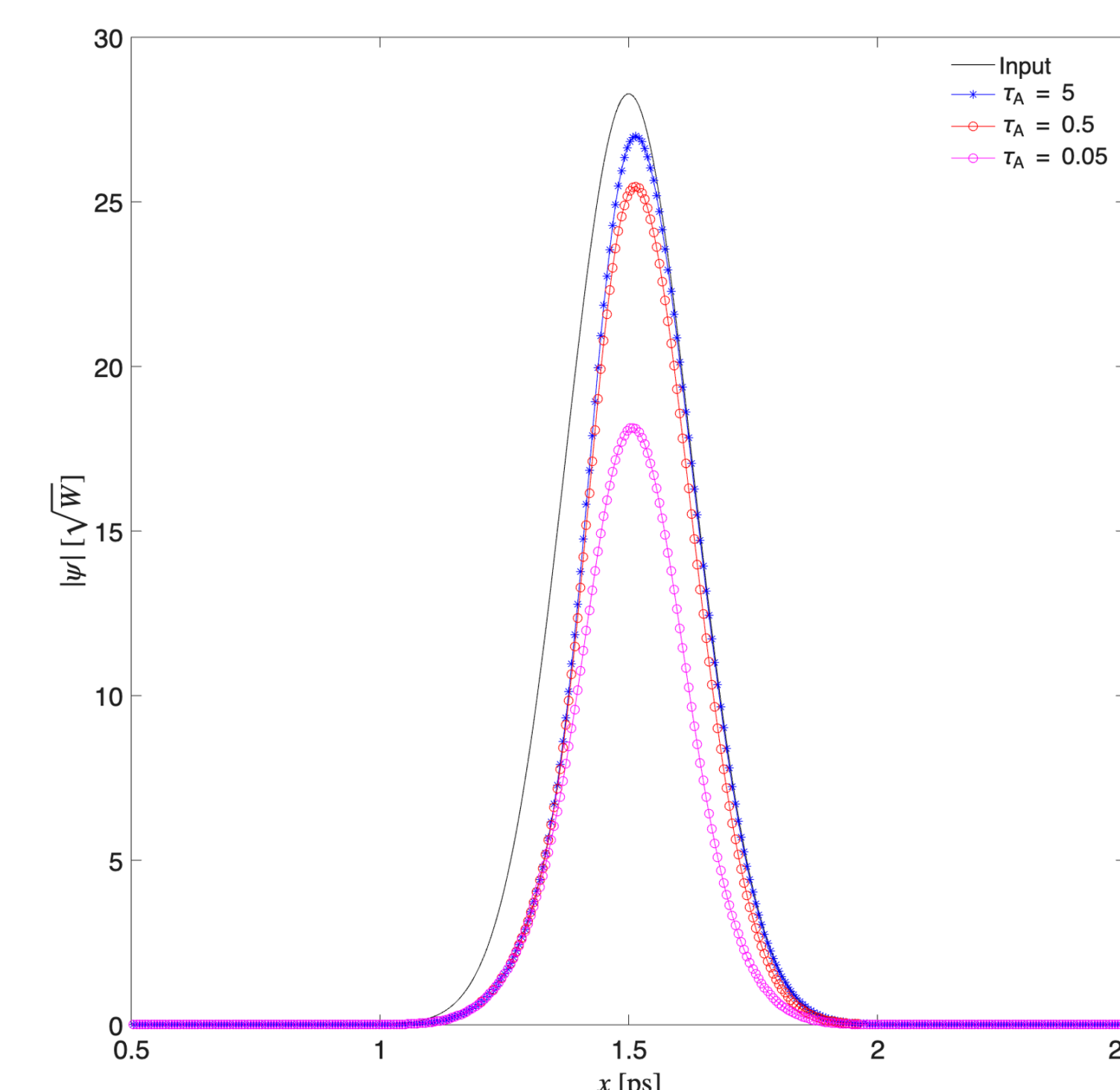


Fig. 3: Pulse output in SESAM comparison for fixed saturation energy and unsaturated loss.

Comparison	T and T^*	$\mathcal{U}^{\text{SESAM}}$ and $(\mathcal{U}^{\text{SESAM}})^*$
Relative Errors	3.1534×10^{-16}	3.1113×10^{-16}

Future work

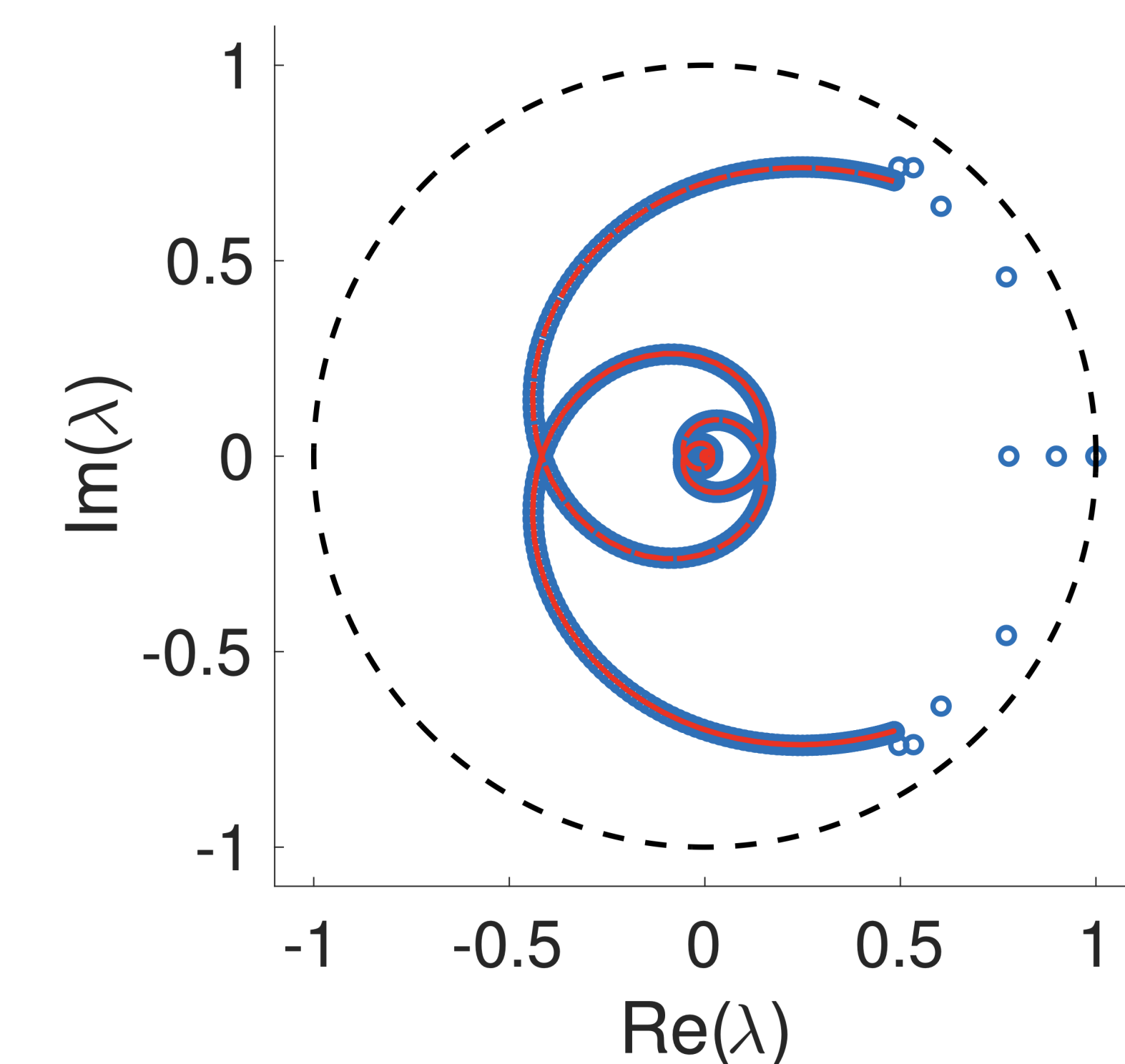


Fig. 4: Spectrum of the monodromy operator for the optimal pulse value using a fast saturable absorber^[2].

- Find breather solutions by optimization.
- Analyze stability of the optimal pulse by computing the spectrum of the corresponding monodromy operator.
- We expect to see new eigenvalues due to noise in the wake of the pulse, which is not suppressed because of the slow response of the SESAM.

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